Mohamed "Oscar" Ghanem

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First-Author Publications

- "The Shape of a Program: Path Signatures for Trace-to-Program Induction." ***NeurIPS 2026***.
- "Flowing Through States: Neural ODE Regularization for Reinforcement Learning." ***ICLR 2026***.
- "Learning Better Representations From Less Data For Propositional Satisfiability." ***NeurIPS 2024 (Spotlight)***.
- "FLoBC: A Decentralized Blockchain-Based Federated Learning Framework." ***IEEE BCCA 2022***.

Employment

Doctoral Researcher February 2024 – Present
CISPA Helmholtz Center

- Research applications of Rough Path Theory to the latent geometry of learned representations (**NeurIPS 2026**).
- Explored applications of neural ODEs as Reinforcement Learning regularizers (**ICLR 2026**).
- Designed various novel neuro-symbolic approaches to problems like SAT and synthesis (**NeurIPS 2024 & 2026**).
- Tutored the Neuro-Symbolic Computing 2024 seminar; reviewer for NeurIPS, ICLR, ICML, & AAAI (2025–2026).

Machine Learning Research Engineer October 2021 – December 2023
SpreeAI

- Developed an automated evaluation suite for Virtual Try-On models with neural KPIs (e.g., structural similarity).
- Built an end-to-end deep learning pipeline for volumetric 3D hair generation from portrait images.
- Implemented and tweaked a state-of-the-art video segmentation model that served most of our machine learning pipelines.
- Undertook many large-scale GAN-based dataset generation processes producing terabytes of training data.
- Collaborated on developing a deep learning pipeline to produce 3D head geometry from portrait images.

Research Assistant June 2022 – April 2023
Max-Planck Institute: MPI-SWS

- Developed a PyTorch implementation for a meta-learning technique that outperformed most benchmarks.
- Built a unifying Python library for all prominent reward-shaping techniques following a **stable-baselines3** style.

Automated Reasoning Tutor November 2021 – February 2022
Max-Planck Institute: MPI-INF

- Conducted weekly tutorials and graded submissions on the foundations of automated theorem provers.

Data Scientist July 2021 – October 2021
Turing.com

- Developed an adaptive time-series model for predicting ad-campaign revenue that remains in production use today.
- Created numerous automated analytics reports for product and marketing teams.

Software Engineer July 2019 – October 2020
Nabta Health Ltd.

- Created numerous React Native components corresponding to four of the main app's primary features.
- Built NFC temperature sensor drivers on both native platforms (Android Java & iOS Obj-C).
- Implemented several REST APIs essential for content-fetching and database synchronization.

Education

Technical University of Munich February 2024 – Present
Munich, Germany

- **PhD in Computer Science**: Machine Learning, *expected* 2028. **Advisor**: Prof. Bernd Finkbeiner.

Saarland University October 2021 – January 2024
Saarbrücken, Germany

- **M.Sc. in Computer Science**, GPA: **1.3** (German scale, 1.0 = best)
- **Graduate Coursework:** Reinforcement Learning, Machine Learning in Cybersecurity, Neuro-Symbolic Computing, Automated Reasoning, Security, Machine Learning for Program Analysis, Automated Debugging
- Completed two **Research Immersion Labs** with Prof. Adish Singla and Prof. Bernd Finkbeiner.

The American University in Cairo September 2016 – June 2021
Cairo, Egypt

- **B.Sc. in Computer Engineering**, GPA: **3.97/4.0** (American scale)
- **B.Sc. Thesis:** *FLoBC: A Decentralized Blockchain-Based Federated Learning Framework*
  - This project offers a collaborative framework for training ML models without sharing user-specific training data.
- **Minor in Mathematics**, GPA: **4.0/4.0**
- **Mathematics Thesis:** *On Theoretical Complexity and Boolean Satisfiability*

Additional Experience and Awards

- **Top Reviewer @ NeurIPS** 2026, **Silver Reviewer @ ICML** 2026.
- **2nd Place Winner**: Dell's C(4F)2 Artificial Intelligence Hackathon 2021 (students track).
- **Contestant**: ACM Arab and African Collegiate Programming Contest (**ACPC**) in the **2018** & **2019** versions.
- **Finalist**: ACM Egyptian Collegiate Programming Contest (**ECPC**) in the **2018** & **2019** versions.
- **Collegiate Award Holder:** Full-tuition PSSF scholarship (115k USD) for outstanding academic performance.

Languages and Technologies

- **Languages:** Python, Lean, C/C++, CUDA C, Rust
- **Tools/Frameworks:** PyTorch, JAX, Docker